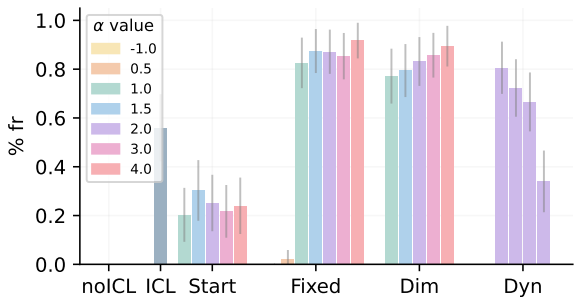
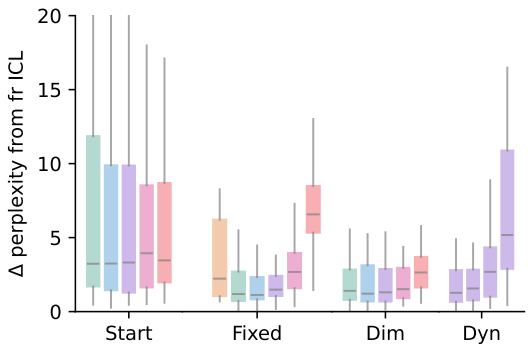
Formal - Informal switch outputs				
Setting	param. α/top-p	*Ask if they wanna go somewhere with you.*	Eval	ppl$(\cdot)$ − ppl(`ICL`)
`noICL`		*I'd be happy to ask them, but I can't make that decision for them.* [...]		
		Δ_{Formal}		
`ICL`		*Inquire as to whether they would care to accompany you.*	Formal	ppl(`ICL`) = `1.23`
`Start`	1.0	*I would suggest asking them directly, "Would you like to go somewh* [...]	Formal	12.59
	1.5	*It would be best to ask them directly if they would like to go som* [...]	Formal	8.71
	2.0	*It would be best to ask them directly if they would like to go som* [...]	Formal	10.02
	3.0	*Upon receiving the request, the AI might respond: "Sure, I'd be ha* [...]	Formal	9.34
	4.0	*Upon entering the room, you notice a group of people huddled aroun* [...]	Formal	10.21
`Fixed`	−1.0	*Wanna go get ice cream? Or sumthin? Or just go for a walk? Or to t* [...]	Formal	266.77
	0.5	*I would be pleased to inquire as to whether they have an inclinati* [...]	Formal	20.15
	1.0	*I would suggest that you approach the individuals in question and* [...]	Formal	36.77
	1.5	*It is inappropriate for me to make such a suggestion to another in* [...]	Formal	47.02
	2.0	*It is inadvisable for an individual to make such a presumption or* [...]	Formal	55.02
	3.0	*Upon having ascertutively ascerted ascertained the disposition to* [...]	Informal	214.77
	4.0	*Upon having having having having having having the occasion inasmu* [...]	Formal	1750.77
`Dim`	1.0	*I would be pleased to inquire as to whether the individuals in que* [...]	Formal	26.27
	1.5	*It is inappropriate for me to make assumptions regarding the desir* [...]	Formal	36.27
	2.0	*It is inappropriate for me, as a grammatically inanimate object, t* [...]	Formal	38.02
	3.0	*Upon having ascertained the requisite degree of familiarity with t* [...]	Formal	83.27
	4.0	*Upon having having having been having the occasion for having been* [...]	Formal	402.77
`Dyn`	0.5	*It is not within the purview of this text to make inquiries on beh* [...]	Formal	21.90
	0.6	*It is not within the purview of this program to engage in hypothet* [...]	Formal	20.77
	0.7	*It is not within the purview of this program to engage in hypothet* [...]	Formal	19.52
	0.95	*I'd be pleased to inquire as to whether or not they would be amena* [...]	Formal	55.02
		$\Delta_{Informal}$		
`ICL`		*Ask them if they'd like to come with you to that place.*	Formal	ppl(`ICL`) = `1.29`
`Start`	1.0	*Wanna go somewhere with ya? Like, where? And why? I'm just askin'* [...]	Informal	2.34
	1.5	*I'd be happy to help you plan a trip! Where would you like to go,* [...]	Formal	2.04
	2.0	*dua' ma'aak (may Allah be with you) and then ask, "Would you like* [...]	Formal	2.80
	3.0	*dua' ma'aakum ila hadha l-ma'a (pronounced dū'ah mā'akum*	Informal	3.12
	4.0	*yawn I'd be happy to accompany you to a location, but I'll need so* [...]	Formal	3.05
`Fixed`	−1.0	*It is advisable to approach the individuals in question and inquir* [...]	Formal	94.71
	0.5	*They wanna go with ya? Let's do it! Where we headed? I'll pack acc* [...]	Informal	2.99
	1.0	*Wanna go on a trip w me? Heck yeah! Where to? I'll pay for all exp* [...]	Formal	11.46
	1.5	*I'm like, "hey, wanna go on a trip? I'll pay for all expenses and* [...]	Formal	13.15
	2.0	*duh, duh's like, "nah, but we'll pay for all expenses. We would li* [...]	Formal	16.21
	3.0	*duppa lil wtf lil*	Informal	83.21
	4.0	*ya kinda kinda kinda like l kinda l kinda kinda lol*	Informal	332.71
`Dim`	1.0	*Wanna go on a trip w me? Heck yeah! Where to? I'm open to suggesti* [...]	Formal	10.34
	1.5	*I'm down if u got a plane or sumthin.*	Informal	13.40
	2.0	*duh, duh's like, hell yeah! but only if i can bring my dog. i'll p* [...]	Informal	14.34
	3.0	*duppa lilpiss n2 get it?*	Informal	30.34
	4.0	*ya kinda like l kinda l kinda lil*	Informal	100.71
`Dyn`	0.5	*They wanna go with ya? Let's go! Where to?*	Informal	3.12
	0.6	*They wanna go with ya? Ask 'em where and when, then pack your bags* [...]	Informal	1.74
	0.7	*They'd love to come with you to that place! Just let them know the* [...]	Formal	1.35
	0.95	*You can ask them, "Would you like to come with me to [location]?"* [...]	Formal	2.57

Table 6: Example generation from the *formal* and *informal* steering techniques. The prompt is the same for every steering and the model is expected to answer with the [Formal and Informal] style. Also, the perplexity difference from the `ICL` output is reported.

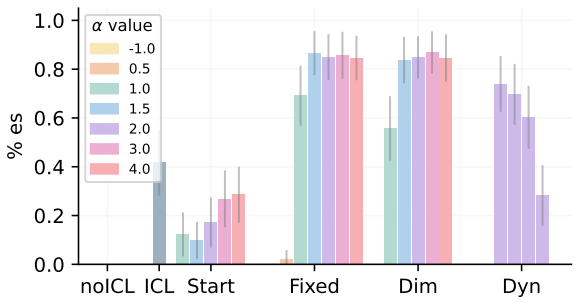


(a) Results for model steering in French

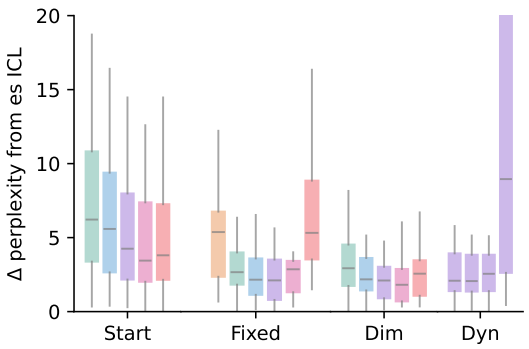


(b) The delta perplexity between different steering techniques calculated w.r.t. the `ICL` generation that follows the reference language (i.e. French)

Figure 7: All techniques proposed toward *French* (fr) steering. The figure includes `Dyn` results with values of $p_{\text{top}} \in [0.5, 0.6, 0.7, 0.9]$ shown in order from left to right.



(a) Results for model steering in Spanish



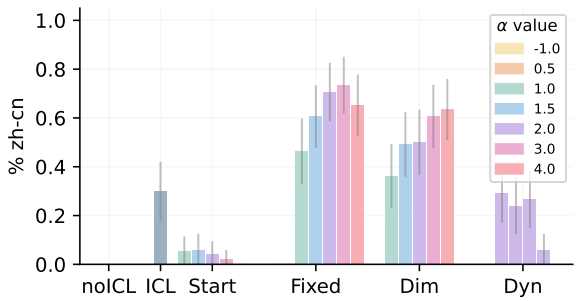
(b) The delta perplexity between different steering techniques calculated w.r.t. the `ICL` generation that follows the reference language (i.e. Spanish)

Figure 8: All techniques proposed toward *Spanish* (es) steering. The figure includes `Dyn` results with values of $p_{\text{top}} \in [0.5, 0.6, 0.7, 0.9]$ shown in order from left to right.

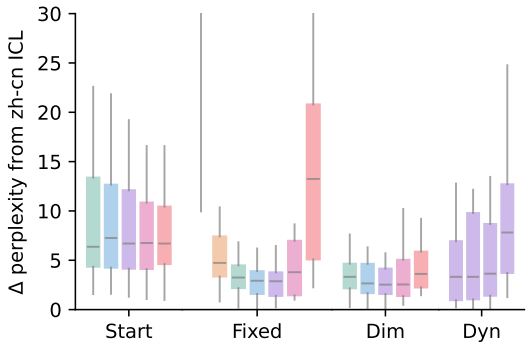
E Steering Vector Insights

Some insights gathered from the steering vectors adopted for the employed tasks are represented below. As per Section A.1 each steering vector has a [layer, head, d_{head}] shape for each generated token. To compress the d_{head} dimension into one single intensity value we used the L_2 norm and the mean in Figures 15a and 15b respectively. As can be seen, there are common patterns among the most important attention heads in terms of intensity, even on different tasks. Furthermore, it can be observed that the attention heads in the last layers tend to play a more important role than those in the first layers. This confirms a pattern known in the literature that has already been observed in the past.

Moreover, it is possible to check how the steering vector intensity changes during the generation process. With this regard, Figure 14 shows, for different generation steps, the L_2 norm of the $\Delta_{Italian}$ steering vector (every other Δ show the same patterns during generation). Some of the most important heads in terms of intensity are consistent during generation, generally lowering their intensity as can be observed from the color bar near each image.

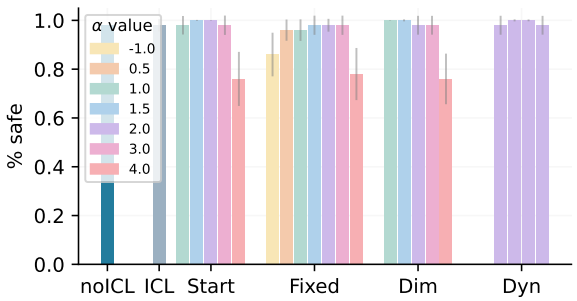


(a) Results for model steering in Chinese

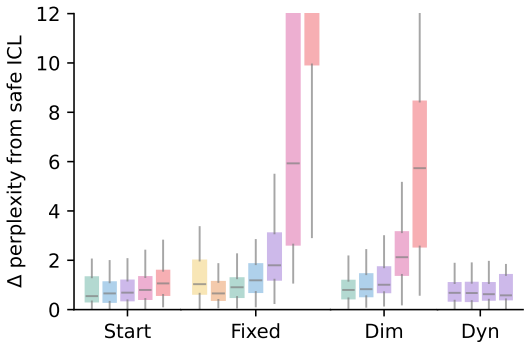


(b) The delta perplexity between different steering techniques calculated w.r.t. the ICL generation that follows the reference language (i.e. Chinese)

Figure 9: All techniques proposed toward *Chinese* (zh-cn) steering. The figure includes Dyn results with values of $p_{\text{top}} \in [0.5, 0.6, 0.7, 0.9]$ shown in order from left to right.



(a) Results for model steering towards safe generation



(b) The delta perplexity between different steering techniques calculated w.r.t. the ICL generation that follows the safe task

Figure 10: All techniques proposed toward *Safe* steering. The figure includes Dyn results with values of $p_{\text{top}} \in [0.5, 0.6, 0.7, 0.9]$ shown in order from left to right.

F Multi-property Steering Results

All the multi-property results are shown in Figure 16 for the *Unsafe* property in combination with all the languages (*Italian*, *French*, *Spanish* and *Chinese*) and, in Figure 17 and 18 for the *formal* and *informal* properties in combination with the *Italian* and *French* languages. For every image, the first row shows the ICL increase from the perplexity of the ICL generation considered as the baseline. The perplexity increase is always counted iff the ICL output includes both properties (e.g. perplexity is calculated iff ICL output is *Unsafe* and *Italian* in the *Unsafe* + *Italian* multi-property test). The second row shows the property performance (*Unsafe*, *Formal* or *Informal*) of the output with all the different settings explored in the main body of the paper. Finally, the third row shows the performance obtained in terms of language from the different settings employed.

G KL Patterns Analysis

It is useful to analyze how the Dyn-amic steering moves during the generation process, to identify some patterns on when stronger intensity is demanded from the steering vector. Table 7 presents some generations following the *Unsafe* property. The steering intensity can be seen as a proxy for the difference in terms of distribution between the original and the steered model. As evident, certain tokens closely related to the property in hand, require more steering than other more general tokens.

Moreover, Table 8 shows different examples from the *Italian* language steering. The highest intensity is mainly focused at the beginning of the generation, providing a strong conditioning about the language to generate. After the first few tokens, the intensity became more sparse, leaving the model almost without intervention and thus achieving better results in terms of perplexity as seen in the results section 7 above.